Parting & Threading

Parting

Parting refers to separating a section of the workpiece, in a similar manner to grooving, by taking a straight cut to a depth equal to the radius of the bar.

Compared to grooving, shallower cuts are made instead, making it less sensitive as the grooves are not as deep, however surface finish, shape, and dimensional accuracy have higher importance and demand more attention.

Usually, the tools for parting/grooving are similar, however the parting insert has an entrance angle, while grooving does not, as grooving needs to have no entrance angle to create proper radial grooves.

As such, the entrance angle of the parting tool causes a pip to form in the middle of the workpiece during parting, however can be prevented by overcutting by 1mm to remove the pip.

If the workpiece is light, to prevent it from flying off the workpiece, the parting tool may be stopped at ~2mm from the center to be manually parted instead.

As the parting tool is also radially fed, it forms clock-spring chips due to all sides of the tool cutting out material. Hence, to prevent jamming, pecking/peck cutting is used to prevent the build up of chips.

Threading

Threads have the following quantities:

ISO threads are quantified by $Ma\ Xb$, where

a is the major diameter,

b is the pitch

0.614b is the pitch depth

a-2(0.614b) is the minor diameter

Threading refers to cutting out external (or even internal threads) to mate with other threaded parts

During threading, feedrate is crucial as it has to match with the pitch, and the coordination per revolution is between the thread pitch (mm/thread) and the feedrate, facilitated by the CNC machine.

Typically, thread turning is done in multiple incremental passes, to prevent overloading on the threading insert, with 6-12 passes done to form the full profile of the thread.

During each pass, there are different ways to infeed the insert.

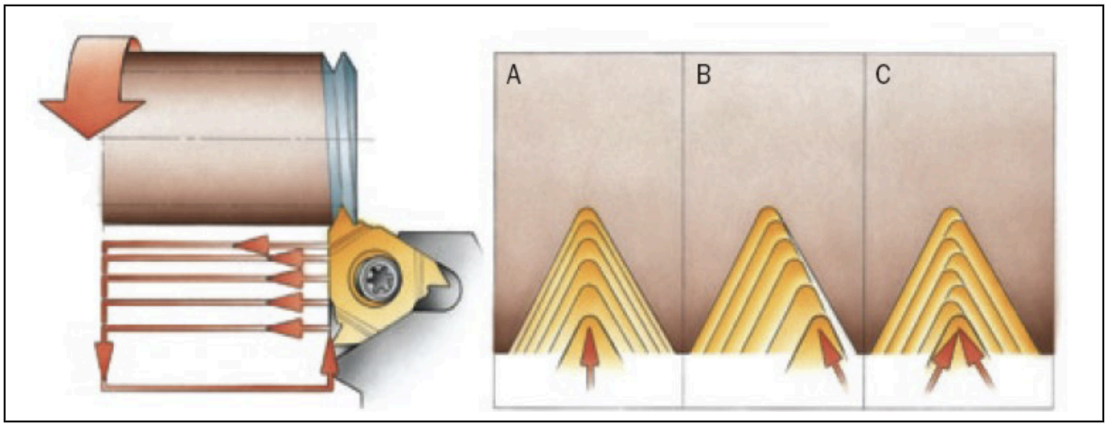

Figure 2: (A) Radial infeed, (B) Modified flank, and (C) Incremental Infeed

Radial infeed: insert is fed radially into work, causing a more even tool wear on both sides of the insert, making it suitable for finer pitches, and is also the most common method

modified flank: insert is fed in the same angle as the profile (~29). Less heat and vibration is generated due to smaller contact, ensuring high production security.

incremental infeed: insert is fed in at opposite sides of the thread profile in alternating steps. Causes more even tool wear and longer tool life, thus best used for coarse or large thread profiles.

For radial infeed, there are two ways to determine the depth of cut for each pass.

Constant depth of cut: the insert is fed in at a constant depth, providing more chip control and longer tool life, but due to the growing area for later passes, causes more cutting forces the insert needs to withstand, causing poor surface finish

Constant area: the insert is fed at decreasing depths, to ensure the same area of work is cut out. the last cut may have no depth of cut, becoming a spring pass, for finishing. This causes better surface finish at the end, but may lower the tool life due to the high depth of cut at the start.

There are also different 2 kinds of threading inserts,

- Full profile threading insert
  - the most common insert, for high productivity as it is fit for only one pitch, and form the full thread profile.
  - They ensure the correct depth, bottom, and top radii to ensure stronger threads, and require no deburring afterwards, and also require fewer passes due to the larger nose radius.
  - However, some extra stock needs to be left (0.03 to 0.07) to top the finishing diameter, and a specific insert is needed for different pitches and profiles.
- Partial profile inserts
  - the insert does not fit the top crests, and has an allowance.
  - used with a minimum tool inventory, as they work with a range of pitches, hence requiring lesser inserts to be bought.
  - they usually have a smaller nose radius as they are designed for smaller pitches, causing lower tool life.
  - When machining external threads, the major diameter must be prepared and controlled prior to threading.
  - When machining internal threads, the minor diameter must be prepared and controlled prior to threading.